

# Lecture 9: Turing Machines

## DFA Impossibility Proof

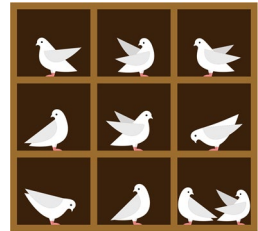
Consider *any* DFA **M**. We show **M** does not decide  $L = \underline{\hspace{2cm}}$

Let  $s =$  \_\_\_\_\_. (It's a finite number!)

Consider the (finite) input string  $\mathbf{x^s y^s}$ .

As we process the string, the DFA switches to some different states  $q_i$ :

|               |           |           |           |           |           |         |           |
|---------------|-----------|-----------|-----------|-----------|-----------|---------|-----------|
| $\varepsilon$ | $x$       | $xx$      | $xxx$     | $xxxx$    | $xxxxx$   | $\dots$ | $x^s$     |
| $q_{i_0}$     | $q_{i_1}$ | $q_{i_2}$ | $q_{i_3}$ | $q_{i_4}$ | $q_{i_5}$ | $\dots$ | $q_{i_s}$ |



There are \_\_\_\_\_ (pigeonholes) and \_\_\_\_\_ (pigeons),  
so there must be \_\_\_\_\_ such that \_\_\_\_\_.

**Claim.** The two inputs \_\_\_\_\_ and \_\_\_\_\_ have the same final state. Why?

This means **M** must

But  $\frac{1}{2} \in L$  and  $\frac{1}{3} \notin L$ , so **M** is wrong on one of these!

**Conclusion:** **M** does not decide L.

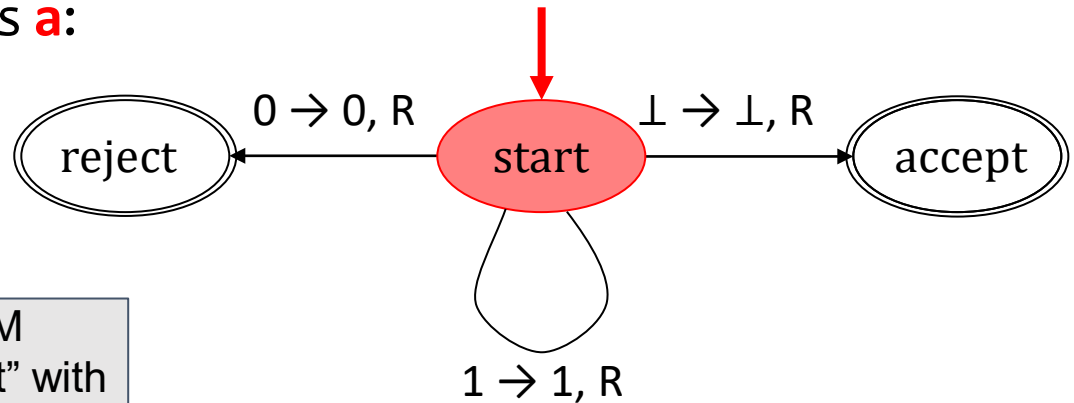
# What is a Turing Machine?

Like a DFA, except each transition also specifies:

- \_\_\_\_\_
- \_\_\_\_\_

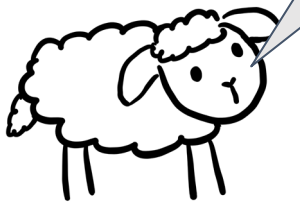
Specifically: “**a  $\rightarrow$  b, R**” means:  
if the current tape cell is **a**:

1. \_\_\_\_\_
2. \_\_\_\_\_

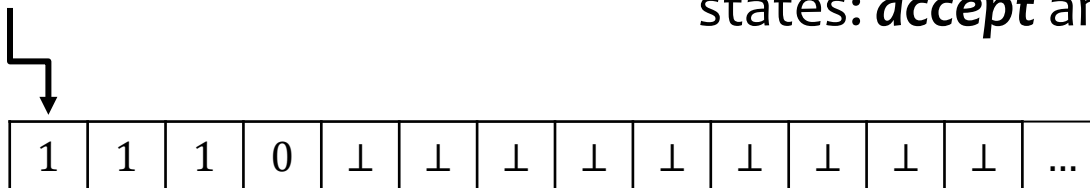


Assume that if TM encounters “move left” with head at position 0, the head doesn’t move

There are two special “termination” states: ***accept*** and ***reject***.



# INFINITE TAPE

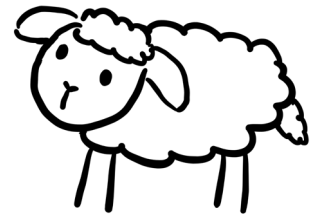


# Formal definition of a Turing Machine

A **Turing Machine** is a 7-tuple:

$$M = (\Sigma, \Gamma, Q, q_{\text{start}}, q_{\text{acc}}, q_{\text{rej}}, \delta)$$

All of these sets are finite



- $\Sigma$  is the (finite) input alphabet,
- $\Gamma$  is the (finite) tape alphabet,  
 $\Sigma \subseteq \Gamma$ , Contains blank symbol  $\perp \in \Gamma$ ,  $\perp \notin \Sigma$
- $Q$  is the *finite* set of states,
  - $q_{\text{start}} \in Q$  is the *initial* state,
  - $q_{\text{acc}}, q_{\text{rej}} \in Q$  are the (distinct) *accept* and *reject* states,  
 $F = \{q_{\text{acc}}, q_{\text{rej}}\}$
- $\delta$  is the *transition function*.

$$\delta: \underbrace{(Q \setminus F)}_{\text{Non-terminal state}} \times \underbrace{\Gamma}_{\text{Cell contents}} \rightarrow \underbrace{Q}_{\text{Next state}} \times \underbrace{\Gamma}_{\text{Symbol to write}} \times \underbrace{\{L, R\}}_{\text{Where to move the head}}$$

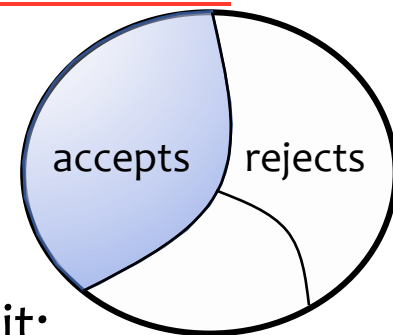
# The Language of a TM

**Question:** What are the possible outcomes of a TM **M** on an input  $x$ ?

**Answer:**  $M(x)$  either (i) accepts, (ii) rejects, or (iii) \_\_\_\_\_

The language of a TM is the set of strings it accepts:

$$L(M) = \{x : M \text{ accepts } x\}$$



**Definition:** A Turing Machine **M** **recognizes** a language  $L$  if it:

1. accepts every string in  $L$ , and
2. \_\_\_\_\_ OR \_\_\_\_\_ for every string not in  $L$ .

**Definition:** A Turing Machine **M** **decides** a language  $L$  if it:

1. accepts every string in  $L$ , and
2. \_\_\_\_\_ every string not in  $L$  (and \_\_\_\_\_)

A language  $L$  is **decidable** if there is a TM that decides  $L$ .

Otherwise  $L$  is **undecidable**.

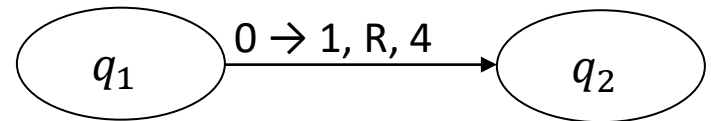
**Fact:** If a language  $L$  is decidable, then the **complement of  $L$**  is also decidable. Why?

**Fact:** If languages  $L_1$  and  $L_2$  are both decidable, then  $L_1 \cap L_2$  is decidable. Why? (can you also prove it for  $L_1 \cup L_2$ ?)

# Tape Jumping TM Equivalence Proof

Claim: A TM that allows the transitions to move the head any distance is exactly as powerful as a standard TM

- E.g.  $\delta(q_1, 0) = (q_2, 1, R, 4)$



- “If in state  $q_1$  and read 0, transition to  $q_2$ , write 1, move 4 spaces right”

Anything a standard TM can do, a tape jumping TM can also do

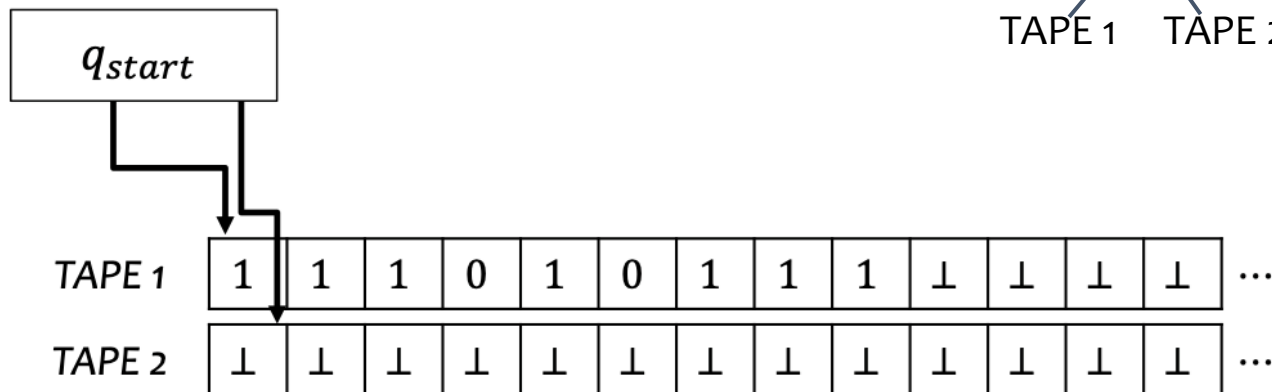
Anything a tape jumping TM can do, a standard TM can also do

# What if we added more tapes to a TM?

Well, the Church-Turing Thesis says there's no "computer" more powerful than a 1-tape TM.

So a 2-tape TM better not be more powerful.  
Let's prove it!

Example transition rule for 2-tape TM:  $\delta(q_1, 1, \perp) = (q_2, 0, 1, R, \dot{L})$



# Equivalence of 1-tape and 2-tape TMs

How can we store the data of 2 tapes if we only have 1 tape?

How do we track where the 2 heads of the tapes are?

What if we need more space on one of the tapes?

How can we know the symbols at 2 separate places?